ECON-423/523-Labs-4-5

Sonan Memon

University of Oregon



Introduction

- As economists, we run a lot of regressions; it is our bread and butter.
- By default, we estimate OLS, which is a linear regression with appealing properties.
- Our goal is to conduct causal inference.

Packages

```
library(tinytex)

library(tidyverse)

library(dslabs)
library(dplyr)

library(ggplot2)

library(tibble)
library(modelsummary)

library(broom)

library(car)

library(fixest)

library(sandwich)
```

OLS 1:

$$y = \beta_0 + \beta_1 * x + e,\ \hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

OLS 1:

```r
set.seed(1)
data <- tibble(x = runif(100,0, 10),
               e = rnorm(100,0, 15),
               y = 10 + 1.5*x + e)

ols_coefficients <- function(x, y) {
  x_bar <- mean(x)
  y_bar <- mean(y)
  beta_1 <- sum((x - x_bar)*(y - y_bar))/sum((x - x_bar)^2)
  beta_0 <- y_bar - beta_1*x_bar
  return(list(intercept = beta_0, slope = beta_1))
}

result <- ols_coefficients(data$x, data$y)

intercept <- result$intercept
slope <- result$slope
```

OLS 1: Scatterplot

```
ggplot(data, aes(x = x, y = y)) +
  geom_point() +
  geom_abline(slope = slope, intercept = intercept)
```

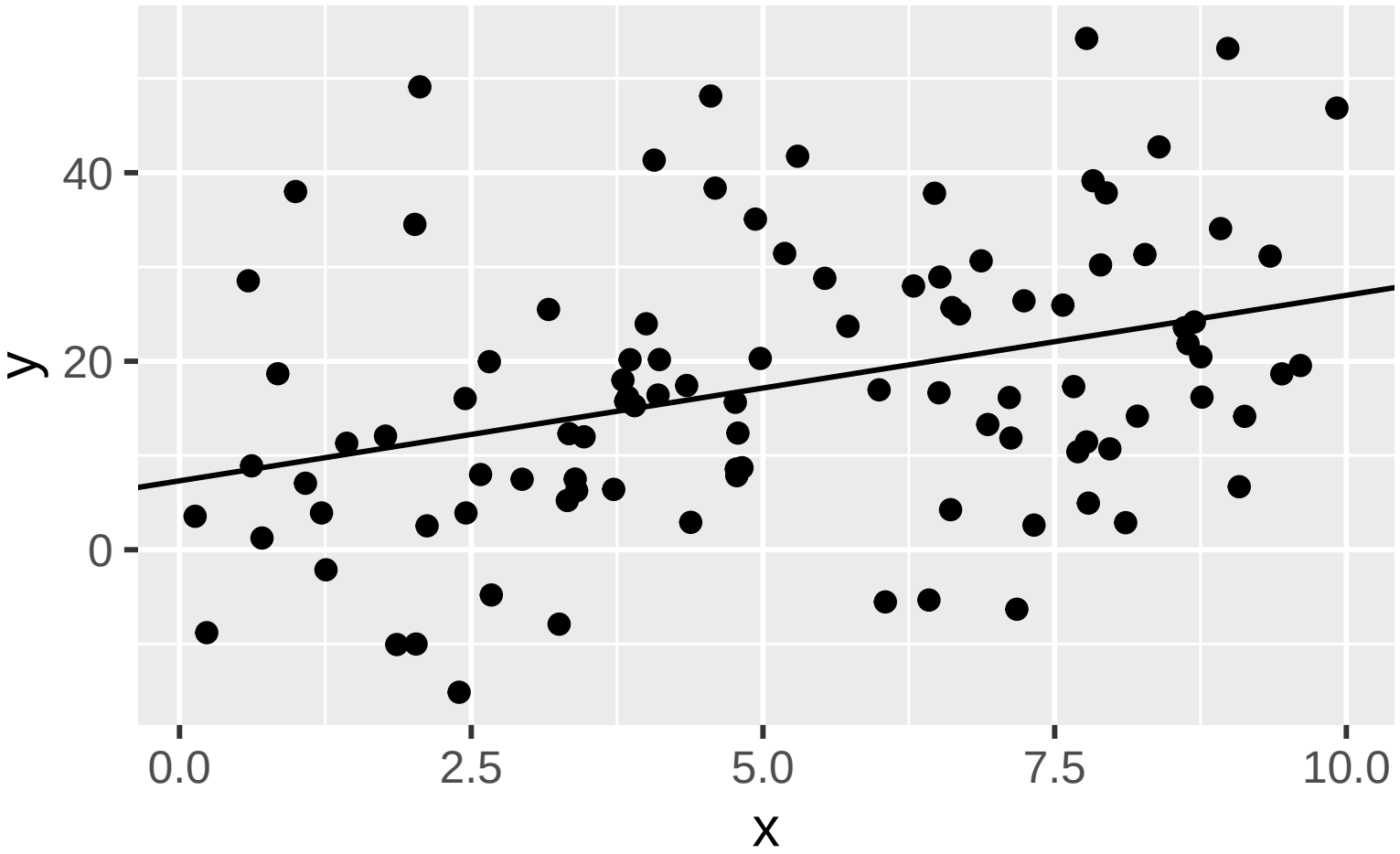


Figure 1: Scatter Plot

OLS 2:

- lm function: for univariate model.

```
result <- lm(data = data, y ~ x)

result$coefficients
```

```
(Intercept)           x
   7.310118    1.968515
```

```
#summary(result)
```

OLS 3:

- lm function: for multivariate case.

```
n = 100
set.seed(1)

data <- tibble(x = runif(n,0, 10),
               z = runif(n, 5, 15),
               e = rnorm(n,0, 15),
               y = 10 + 1.5*x - 3*z + e)


model1 <- lm(data = data, y ~ x)

model2 <- lm(data = data, y ~ x + z)
```

Regression Table

Table 1: Results From Simulated Data

	Regression 1	Regression 2
(Intercept)	-20.286***	9.752
	(3.625)	(6.275)
x	1.345*	1.396*
	(0.622)	(0.545)
z		-2.978***
		(0.537)

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 2: Simulated Data-1

Confidence Interval

```
n = 100

set.seed(1)

data = tibble::tibble(
  e = rnorm(n, sd = 2),
  v = rnorm(n, sd = 1),
  x = runif(n, min = 0, max = 10),
  y = 8 - 3 * x + e,
  z = 20 - 0.3 * y + 3 * x + v
)


lm1 <-lm(data = data, z~y)
lm2 <- lm(data = data, z~y + x)
```

Confidence Interval

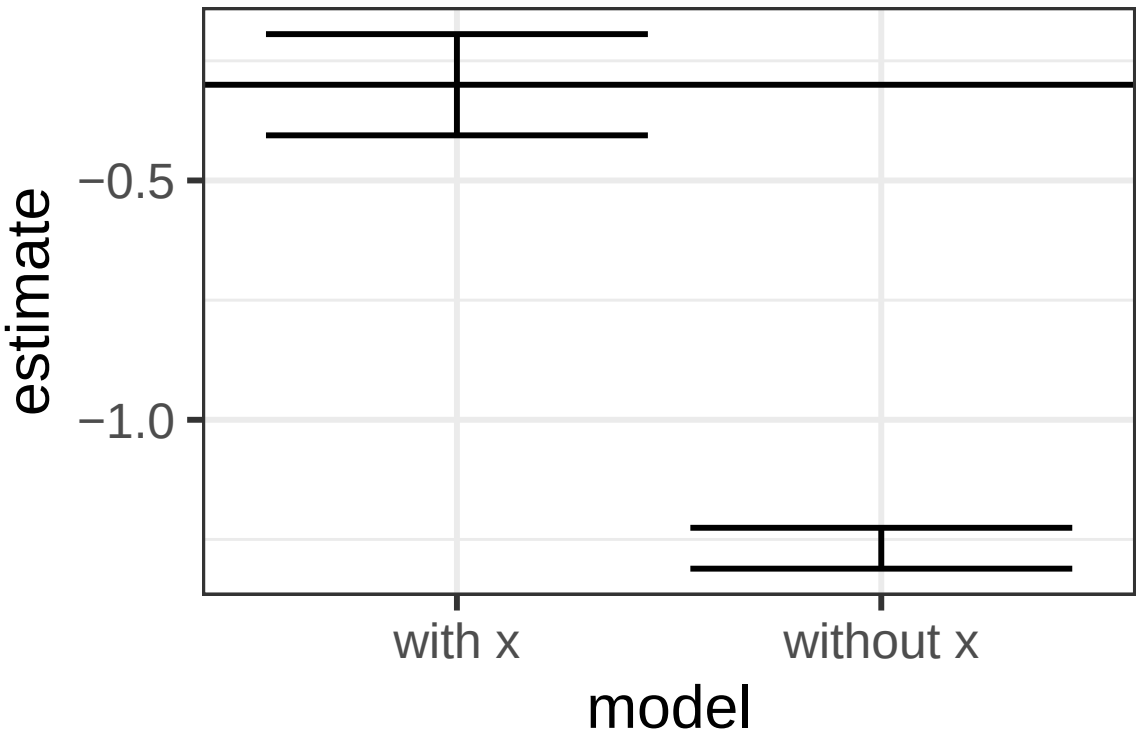


Figure 3: Confidence Bands

Hypothesis Testing:

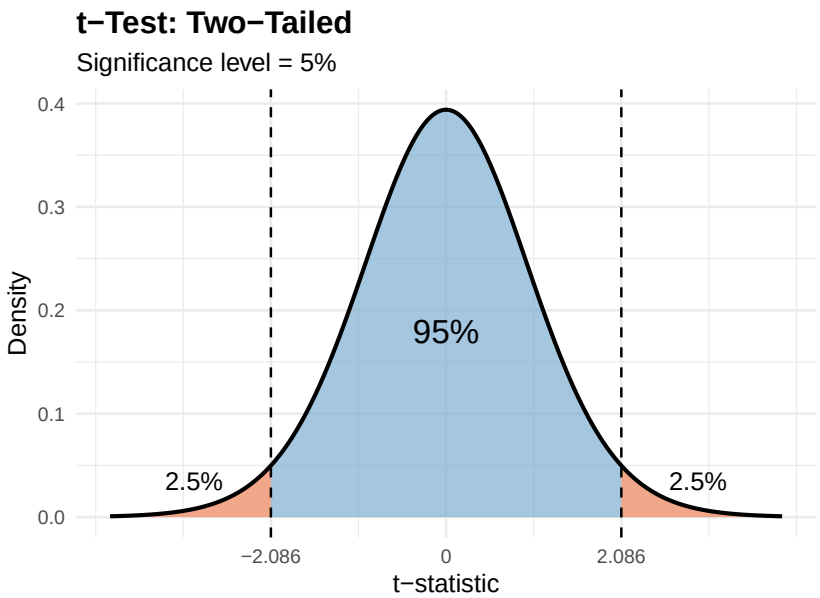
- $$H_0 : \beta = 0.5 \;\; H_a : \beta \neq 0.5$$
- Test statistic:
$$\frac{\hat{\beta} - \beta}{s.e}$$
- Joint hypothesis test for all slope coefficients

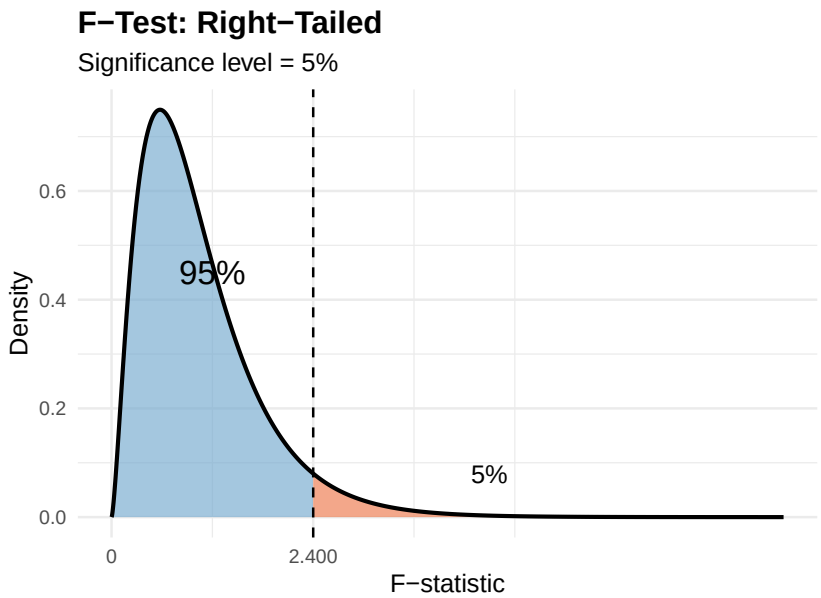
$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

$$H_a : \text{At least one } \beta_j \neq 0$$

$$F = \frac{(SSR_R - SSR_{UR})/q}{SSR_{UR}/(n-k-1)}$$

Hypothesis Testing: t and F Distributions





Hypothesis Testing:

```
beta1 = results$estimate[1]
st.err1 = results$std.error[1]

test1 = (beta1 - (-0.3))/st.err1
test1
```

```
[1] -44.40833
```

```
beta2 = results$estimate[2]
st.err2 = results$std.error[2]

test2 = (beta2 - (-0.3))/st.err2
test2
```

```
[1] 0.001330759
```

Ifelse Again:

```
ifelse(abs(test1) > 1.96,
"Reject the null hypothesis",
"Fail to reject the null hypothesis")
```

```
[1] "Reject the null hypothesis"
```

Hypothesis Testing-1:

```
library(car)


linearHypothesis(lm1, "y = -0.3")


linearHypothesis(lm2, "y = -0.3")
```

Hypothesis Testing-2:

```r
data <- readRDS("running_data.rds")

model_full <- lm(
  time_5k ~ miles_week + sleep_hours +
    resting_hr + age + cross_train + temp_outdoor,
  data = data
)



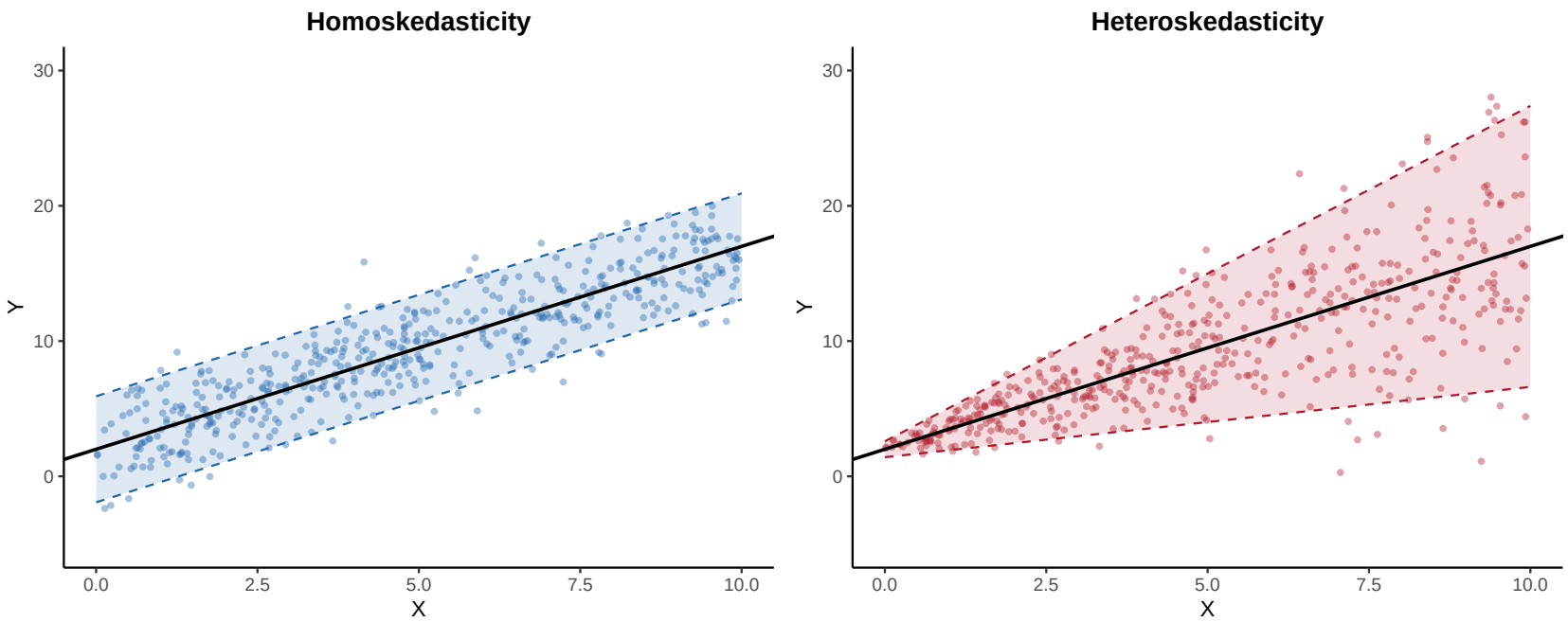
linearHypothesis(
  model_full,
  c(
    "miles_week = 0","sleep_hours = 0",
    "resting_hr = 0","age = 0", "cross_train = 0",
    "temp_outdoor = 0"
  )
)
```

Regression with Interactions:

```
model_interaction <- lm(
  time_5k ~ miles_week * temp_outdoor +
    sleep_hours + resting_hr + age + cross_train,
  data = data
)

m <- feols( time_5k ~ miles_week * temp_outdoor +
    sleep_hours + resting_hr + age + cross_train,
  data = data)
```

Heteroskedasticity:



FEOLS: Heteroskedasticity

```
library(fixest)

data <- read.csv("nlsy79-2020.csv")


m1 <- feols(log_wealth ~ foreign_language, data = data)

m2 <- feols(log_wealth ~ foreign_language + sex + log_educ,
            data = data, vcov = "hetero")

m2 <- lm(
  log_wealth ~ foreign_language + sex + log_educ,
  data = data
)

# Heteroskedasticity-robust standard errors:

coeftest(m2, vcov. = vcovHC(m2, type = "HC1"))
```

FEOLS: Sub-Sets and Sample Weights

```
library(fixest)


m3 <- feols(
  log_wealth ~ foreign_language + sex,
  data = data,
  subset = ~log_wealth >= 10
)

# Inverse, Probability Weights, Frequency
# Weights, Precision Weights:

#m_weights <- feols(
#log_wealth ~ foreign_language + sex,
 # data = data,
#weights = ~sample_weight
#)
```

FEOLS: Clustering of Standard Errors

```
library(fixest)

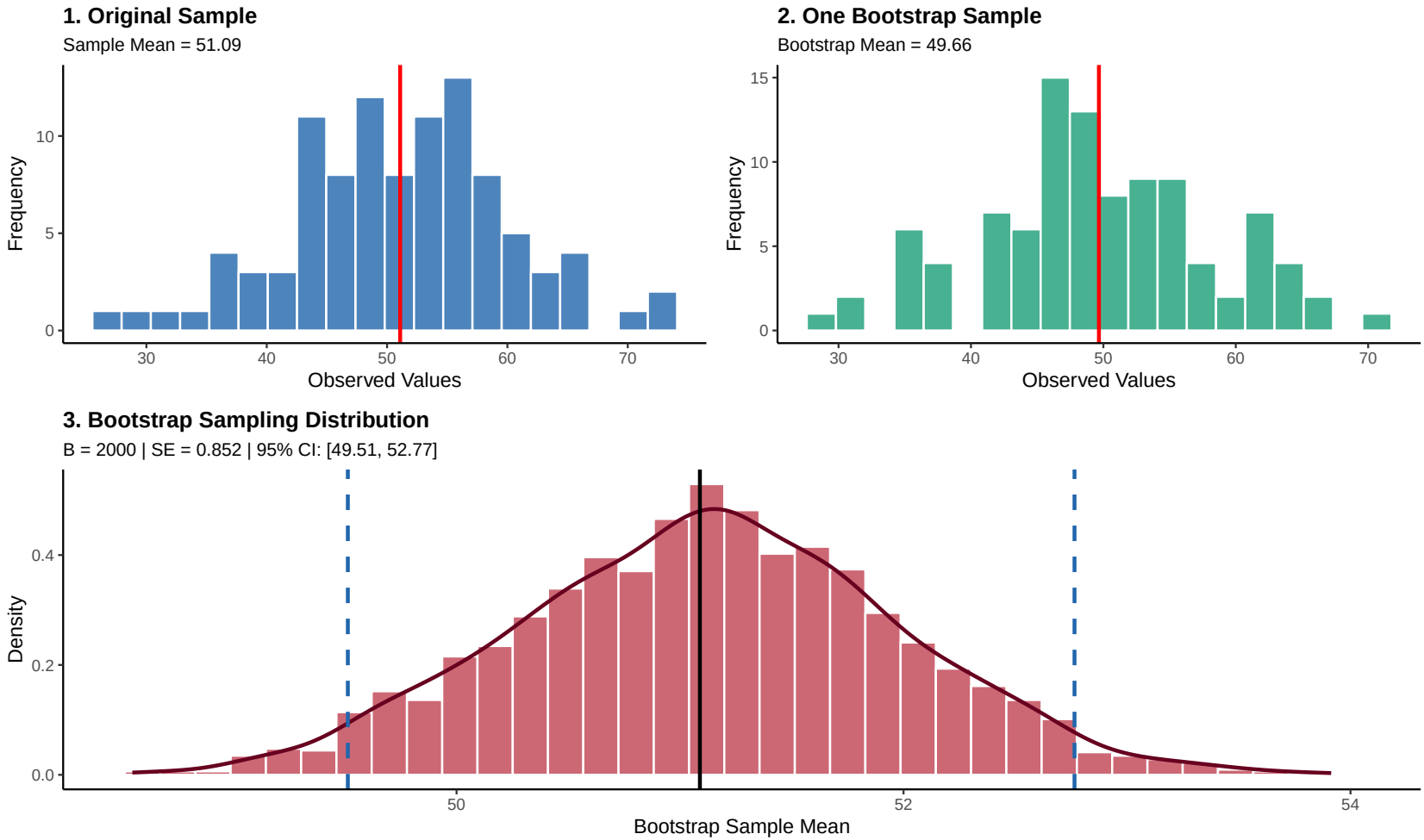
# Clustered Standard Errors:


#m4 <- feols(log_wealth ~ foreign_language + sex + #log_educ,
  #data = data, vcov = ~county



#m_panel <- feols(
  #wage ~ education + experience | person_id + year,
  #data = data,
  #vcov = ~ person_id + year
#)
```

Bootstrapping: Graphical Illustration



OLS vs. Robust vs. Bootstrap Inference

